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Thirsty for power and water, AI-crunching data centers sprout across the West

With promises of jobs and hopes for tax breaks, server farms are reshaping local grids, plumbing, and politics. Are they a boon for communities, or a burden?

By **Felicity Barringer < https://andthewest.stanford.edu/author/felicity-barringer/>**

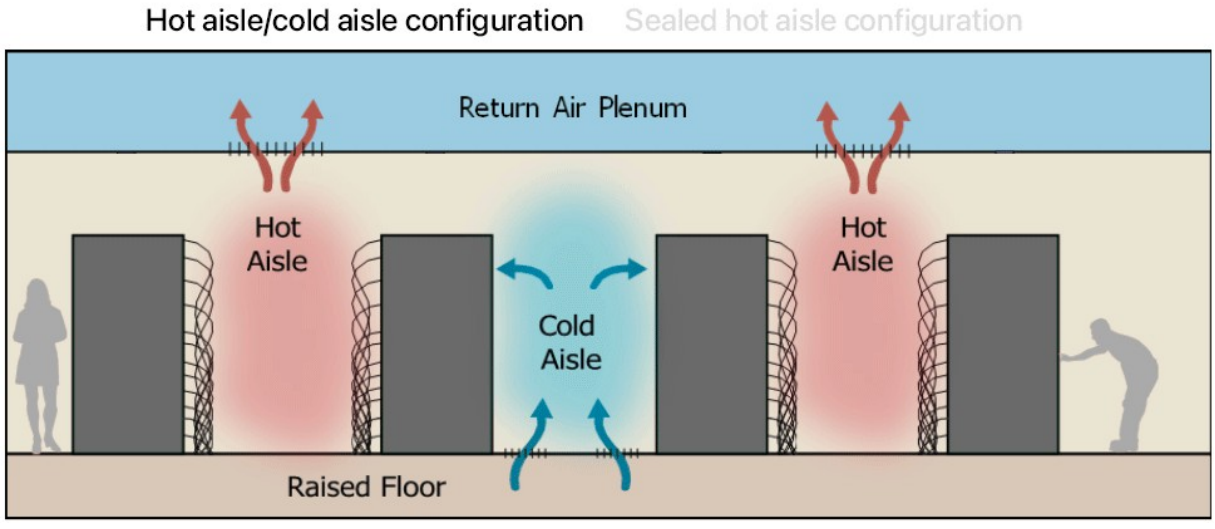
April 8, 2025 < https://andthewest.stanford.edu/2025/thirsty-for-power-and-water-ai-crunching-data-centers-sprout-across-the-west/>

Beans to bits The agricultural community of Quincy, Washington, population 6,220, approved a Microsoft data center in 2006. The tech giant sought room to grow and cheap, abundant electricity from nearby Columbia River hydropower. The facility's opening saw the owners presented with beans from the last crop harvested on the site. TEDDER VIA **WIKIMEDIA COMMONS < HTTPS://COMMONS.WIKIMEDIA.ORG/WIKI/FILE:QUINCY_WASHINGTON_-_AERIAL.JPG>**

By Felicity Barringer

Driving around the cities and small towns of the West, one of the most consequential changes to the landscape are hard to see. Data centers, the buildings of the future, are usually low-slung, their large bulk is best seen from above. A drone's-eye view shows a spreading, warehouse-flat landscape born of the economic and electrical revolution that is reshaping places like Phoenix, the city of Santa Clara in Silicon Valley, or rural Oregon towns close to the Columbia River.

Data centers are remaking local communities and economies around the country, most dramatically in the northern Virginia suburbs of Washington, D.C., but also around the West. These centers, the beating hearts of the internet, cloud computing, and artificial intelligence, sprawl over tens or hundreds of thousands of square feet. At their core are halls filled with identical rows of hundreds of computer servers arranged in aisles – a "cold aisle" where the server draws in cool air, and a "hot aisle" where exhaust is vented.



US Department of Energy

Heat is the enemy of data operations, reducing their efficiency or even making them inoperable. What creates the heat?

The armies of servers gobbling up vast amounts of electricity. What cools it? A variety of technologies, with one, evaporative cooling, requiring significant amounts of water.

This onrushing electronic future means that areas of the West seeking to accumulate data centers need more and more electricity.

This onrushing electronic future means that areas of the West seeking to accumulate data centers need more and more electricity. The growth of artificial intelligence, which uses larger and more complex chips needing far more power, only accelerates the power demand. More transmission lines are needed to get the power to the centers. The share of states' and communities' energy consumed by data centers has grown dramatically: In Arizona, they use 7.4 percent of the state's power, in Oregon 11.4 percent, according to **Visual Capitalist < https://www.visualcapitalist.com/mapped-data-center-electricity-consumption-by-state/>** .

Data centers in Quincy, Washington



Sources: ***Open Street Map*** *(data centers); ESRI (aerial imagery);* & the West

It's not just electricity that is needed by these centers, which are now growing to brobdingnagian sizes to accommodate AI's computing demands. They need significant amounts of water to keep cool, and new ones will need more. The new computer chips that are

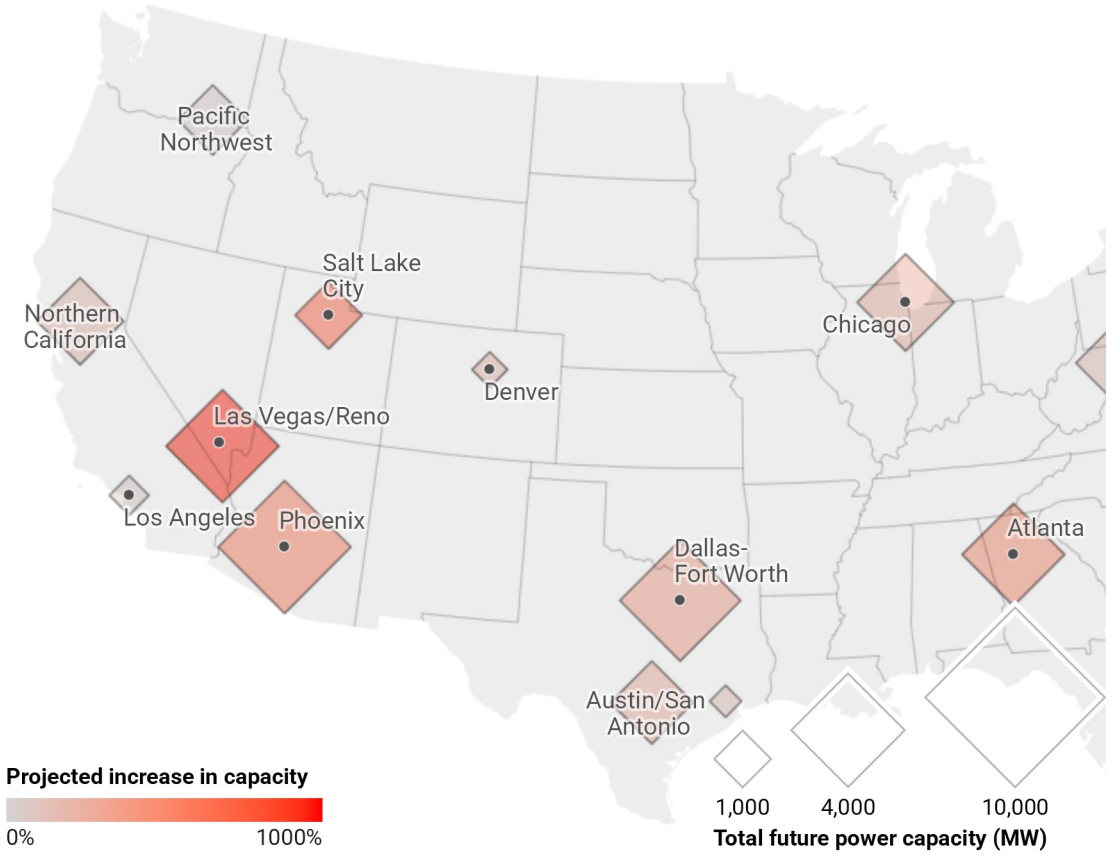
designed to produce artificial intelligence, like Nvidia's Blackwell chip, use far more electricity and therefore need more cooling than the older generation of chips. Nvidia is based in Santa Clara.

Red Canary magazine **recently reported < https://redcanarycollective.org/magazine/phoenix-data-center-water-usage/#:~:text=Assuming%20that%2059%20of%20Maricopa,to%20cool%20data%20centers%20daily>** that the water demand of nearly 60 centers in Phoenix is about 177 million gallons a day – significant, but a fraction of the water used in local agriculture.

All told, the centers are reshaping local grids, plumbing, and politics.

Data Center Hot Spots in the U.S.

Areas with fastest data-center growth and planned capacity



Map: & the West • Source: Upwind, JLL 2024 Data Center Report • Get the data • Download image • Created with Datawrapper

Future planned capacity relative to current capacity. Western areas are **in bold**

	Percent projected growth	Future power capacity (MW)	Number of equivalent households that could be powered

Las Vegas / Reno	953	3,812	3,136,731
Salt Lake City	699	1,271	1,045,851
Phoenix	554	5,340	4,394,057
Atlanta	484	3,125	2,571,429
Dallas-Fort Worth	355	4,396	3,617,280
Chicago	239	2,773	2,281,783
Austin/San Antonio	237	1,993	1,639,954
Denver	196	311	255,909
Northern California	172	2,204	1,813,577
Northern Virginia	142	11,077	9,114,789
Houston	109	244	200,777
Los Angeles	50	398	327,497
Pacific Northwest	40	1,385	1,139,657
New York	39	264	217,234
New Jersey	37	781	642,651
Boston	0	160	131,657

Chart: & the West • Source: EIA and JLL via Upwind • Get the data • Download image • Created with Datawrapper

The economic value proposition for rural areas

"If you ask how many more data centers we need – the answer is always more," said Kevin Park, a Santa Clara city councilman whose support for the projects recently turned to opposition. In populous areas like Santa Clara and Phoenix, the data-center revolution, strongly supported by the labor unions whose workers build the facilities, is becoming more problematic for local officials.

Economically-challenged rural areas, by comparison, tend to welcome them. In Oregon, companies like Meta, Apple, Twitter and Google's parent company have received hundreds of millions of dollars in tax breaks to **entice date-centers < https://www.oregonlive.com/silicon-forest/2023/02/here-are-the-companies-collecting-oregons-b%20iggest-tax-breaks.html>** . **Morrow County is about 160 miles east of Portland on the Columbia River < https://www.oregonlive.com/silicon-forest/2023/05/amazon-secures-1-billion-in-tax-breaks-for-five-new-data-centers-in-eastern-oregon.html>** . Two years ago it **awarded Amazon < https://www.oregonlive.com/silicon-forest/2023/05/amazon-secures-1-billion-in-tax-breaks-for-five-new-data-centers-in-eastern-oregon.html>** nearly $1 billion in breaks over 15 years to build five new data centers.

For some in Oregon, the economic advantages far outweigh the environmental costs. Mark Morgan, the assistant city manager in Hermiston, Oregon, said, "Data centers filled a really

good niche in our economy.... It's a tough nut to have growth and a small-town feel," which residents say they want. The jobs issue, he said, is distorted by big-city media outlets that don't understand rural economies. "Look at the hayseeds in Hermiston – it's a data center that only employs 200 people....

"There's more nuance to it," he continued. "Even setting aside whether we could attract an Intel chip fab that has 10,000 jobs – even if that was possible, the community doesn't want to be overwhelmed with 10,000 jobs. Data centers with 200 jobs end up fitting in pretty nicely."

The greater Hermiston region in Umatilla County has about 10 data centers. Some of the town's other jobs involve processing one of the main crops, potatoes. "Those types of agricultural processing jobs turning potatoes into chips and fries are dangerous and pay extremely poor wages," he said.

> "The community doesn't want to be overwhelmed with 10,000 jobs. Data centers with 200 jobs end up fitting in pretty nicely."
>
> – Mark Morgan, assistant city manager, Hermiston, Oregon

In Oregon's Morrow County, just west of Umatilla County the economy took repeated blows in recent decades when **timber activity withered < https://www.oregon.gov/odf/Documents/workingforests/oregonstimberharvests.pdf>** and **a local coal plant closed < https://portlandgeneral.com/news/2020-10-15-portland-general-electric-announces-end-to-coal-fired-power>** . "You've got to have economic activity or your town will die," said David Sykes, a county commissioner. He has welcomed both **the Amazon data centers < https://portlandgeneral.com/news/2020-10-15-portland-general-electric-announces-end-to-coal-fired-power>** built over the past 15 years and the **new ones on the horizon < https://portlandgeneral.com/news/2020-10-15-portland-general-electric-announces-end-to-coal-fired-power>** . Now, "Young people are coming back. Data centers and wind and solar support school bonds and health districts."

Not everyone agrees. Morrow County voters' unsuccessful **attempt to recall < https://www.opb.org/article/2024/07/22/morrow-county-will-hold-vote-recall-commissioners/#:~:text=The%202022%20recall%20resulted%20in,her%20reelection%20bid%20to%20Wenholz.>** Sykes and two of his commission colleagues last year was based, in part, on the tax breaks they had voted to give Amazon, arguing that Amazon should be sending more in-lieu-of-tax payments back to the community.

Tax breaks were granted several times over the last 15 years. The cumulative impact last year in Morrow County meant that, according to a tax assessor's spreadsheet, Amazon avoided $83,877,339 in taxes but did pay $14,503,485.07 in fees and other contributions. Five years before, Amazon's tax exemptions were $1,975,338.43, and were outweighed by fees and other

payments of $2,346.360. But the value of the exemptions multiplied several times over in 2021 and 2022 before reaching the 2024 total.

Data centers in Boardman, Oregon



Sources: ***Open Street Map*** *(data centers); ESRI (aerial imagery);* & the West

Here's how Morgan, the assistant city manager, thinks about **these tax breaks < https://www.datacenterknowledge.com/hyperscalers/report-amazon-gets-tax-breaks-to-build-more-oregon-data-centers>** : "It didn't cost us anything if they weren't here in the first place." Here's his metric: "Look at who are the largest taxpayers. How many employees do they have? ...The hybrid corn seed facility Corteva [once a division of DuPont] was paying about $2,400 per year per employee.... The WalMart distribution center, about $800 per employee... When I add [data center] payments in lieu of taxes coming to Hermiston and Umatilla County – if you assume a center has 200 jobs, they are paying $40,000 per employee."

If Oregon and its counties offer data-center companies property tax breaks to locate there, other states have different incentives. Arizona, for instance, offers **sales tax breaks on data center equipment < https://www.streamdatacenters.com/resource-library/glossary/arizona-state-data-center-sales-tax-incentives/>** .

Silicon Valley's urban cluster absorbing the most data centers

For data centers, the attraction of Santa Clara, an industrial city in Silicon Valley that was once a chip-making hub, is not tax breaks but cheaper power. The municipal utility, Silicon Valley Power, charges less than Pacific Gas & Electric, the large regional utility. The city now houses about 60 data centers, up from a handful two decades ago. The city council just **approved < https://www.svvoice.com/another-data-center-coming-to-santa-clara/>** another one to be operated by **Prime Data Centers < https://primedatacenters.com/>** , a Connecticut company.

Data centers in Santa Clara, California



Sources: ***Open Street Map*** *(data centers); ESRI (aerial imagery);* & the West

The December vote was six to one. Kevin Park opposed a data center for the first time. “No, I didn’t change my mind,” said Park. “I’ve given people an indication of where we’re going. Now we’re here.”

The quick multiplication of data centers has taxed Silicon Valley Power’s ability to serve them. Kathleen Hughes, an assistant director of SVP, said in an interview that from 2002 to 2024, the company’s electrical power sales rose from 2,400 gigawatt hours to 4,500 gigawatt hours. Data centers use about 60 percent of that power.





Data centers and supporting infrastructure in Santa Clara (clockwise from top left): data center at 1111 Comstock Street; Silicon Valley Power's Agnes Road electrical substation; Comstock Street data center; one of two Core Site data centers on Stender Way; Amazon data center on Mission College Boulevard; *Felicity Barringer/& the West*

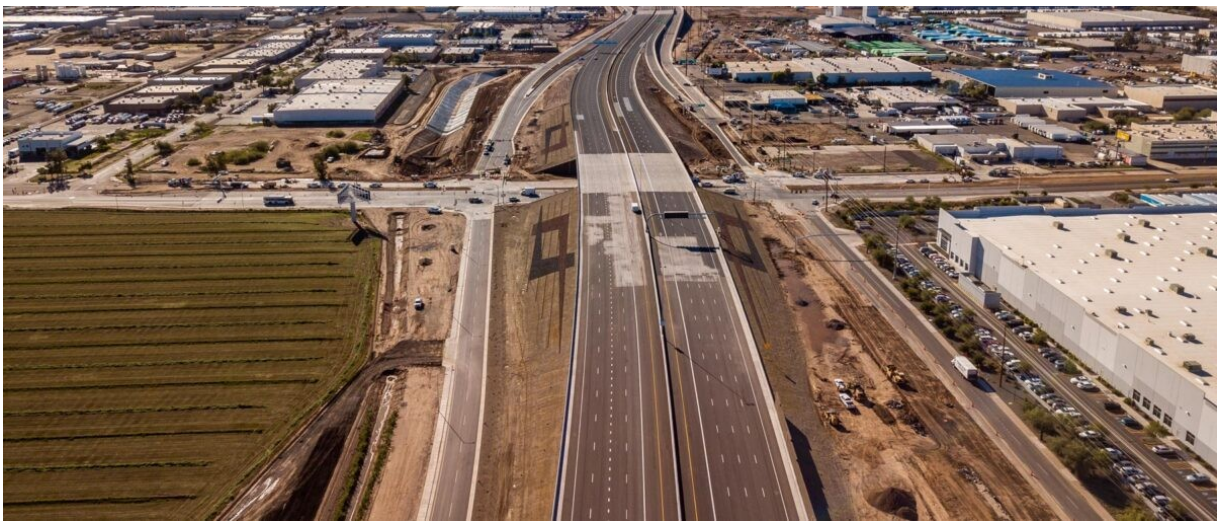
The city's population has also steadily increased, rising from 102,000 in 2020 to 133,000 today. Add state mandates more housing construction, it becomes harder to keep residential and industrial areas separate. As Park said, "In the past, [there were] clearly delineated and clearly defined light industrial areas and residential areas. All those lines are blurring."

> "I consider data centers to be the tapeworms of the city."
>
> – Lance Saleme, Santa Clara planning commission chair

The two recent council approvals overrode the planning commission's objections. As Lance Saleme, the planning commission chair, **said acidly < https://www.azcentral.com/story/news/local/arizona/2021/08/12/2020-census-phoenix-now-fastest-growing-big-city-u-s/8108506002/>** after a January meeting, "I consider data centers to be the tapeworms of the city." In a later interview he explained: A center "is unto itself, its own entity. There's little it will add to the community.... it consumes, it grows, it uses resources. It doesn't kill you, but it doesn't make you healthy."

Fitting data centers into a growing Phoenix area is getting harder





In 2019, the newly dedicated South Mountain Freeway passed through the growing Phoenix community of Laveen. Late last year, Amazon paid $277 million to buy 220 open acres in Laveen for a new data center. *Arizona Department of Transportation via* ***Flickr < https://flickr.com/photos/arizonadot/49240079436/in/photolist-2i2bh7Q-2i4tFeL-2i4tEDY-2i4tG9g-2i4uE5R-2i4uEGT-2i4ri6d-2i4uDdq-2i4tENv-2i4uDyR-2i4uE86-2i4rhLq-2i4rhFA-2i4tERm-2i2coHV-2i4uAAb-2i4uDEh-2i28QLW-2i4uDH3-2i2bhdB-2i4tEUh-2i2coFF-2i4tFhr-2i4tFFN-2i2cp2F-2i2bhx4-2i4riuz-2hNwqWG-2i4riix-2i2coMs-2i4tFU8-2i4tFnX-2i4riHA-2i4ribd-2i2coXx-2hNwqZs-2hNtPZU-2i4tFYr-2i4ripV-2i4uEyg-2i4uEhp-2i4rizE-2hNtPUo-2i4tGhc-2i4uERW-2i2coSh-2i4uELA-2i28Rds-2hNxrGJ-2i4rjio>***

In the Phoenix area, population growth has moved much faster than in Santa Clara. Phoenix was the **fastest-growing big city < https://www.azcentral.com/story/news/local/arizona/2021/08/12/2020-census-phoenix-now-fastest-growing-big-city-u-s/8108506002/>** in the country from 2010 to 2020; it is now the nation's fifth largest, with 1.6 million residents. Around it, Maricopa County has the **second-largest concentration < https://azbigmedia.com/real-estate/phoenix-ranks-as-the-second-largest-data-center-market-in-the-u-s/>** of data centers. Part of the region's appeal is its isolation from most natural disasters, other than heat.

Phoenix is also located **on a major fiber-optic pipeline < https://www.apmresearchlab.org/10x/data-centers-resource>** . Like Santa Clara, it has a government-owned local power utility, **the Salt River Project < https://www.srpnet.com/about/about-srp>** . The operations of Phoenix data centers draw 1.5 gigawatts of power, according to **a 2024 report < https://cushwake.cld.bz/Americas-Data-Center-H2-2024-Update/2/>** from the real estate firm Cushman & Wakefield. (Demand is measured in watts consumed, reflecting both the number of servers in a center and power used for storage.)

Power demands increase fast. The centers' employment rises slowly. The average 12-megawatt center built a decade ago had **20 to 22**

employees < https://www.washingtonpost.com/technology/2024/09/17/data-center-workers-jobs/> doing maintenance, technical and security jobs. Phoenix officials prefer to see the rise in population matched by a rise in available jobs. As Mayor Kate Gallego said in an interview, "In approving new development...we want to prioritize high-wage employment and employers who are involved in the community."

> "A data center takes a lot of land but they don't provide enough jobs for the infrastructure investment."
>
> – Alan Stephenson, Phoenix deputy city manager

As the farm economy shrinks, leaving fallow land, Phoenix has large parcels available, ideal for large new data centers. City officials have envisioned the shape of the city going forward; new urban villages combining housing, jobs, and public transportation networks < https://www.phoenix.gov/pddsite/Documents/PZ/pdd_pz_pdf_00109.pdf> . As Mayor Gallego said, "The city of Phoenix designated employment corridors where we want to see high-wage high-quality jobs." State law requires the city to make that happen without zoning changes that devalue land.

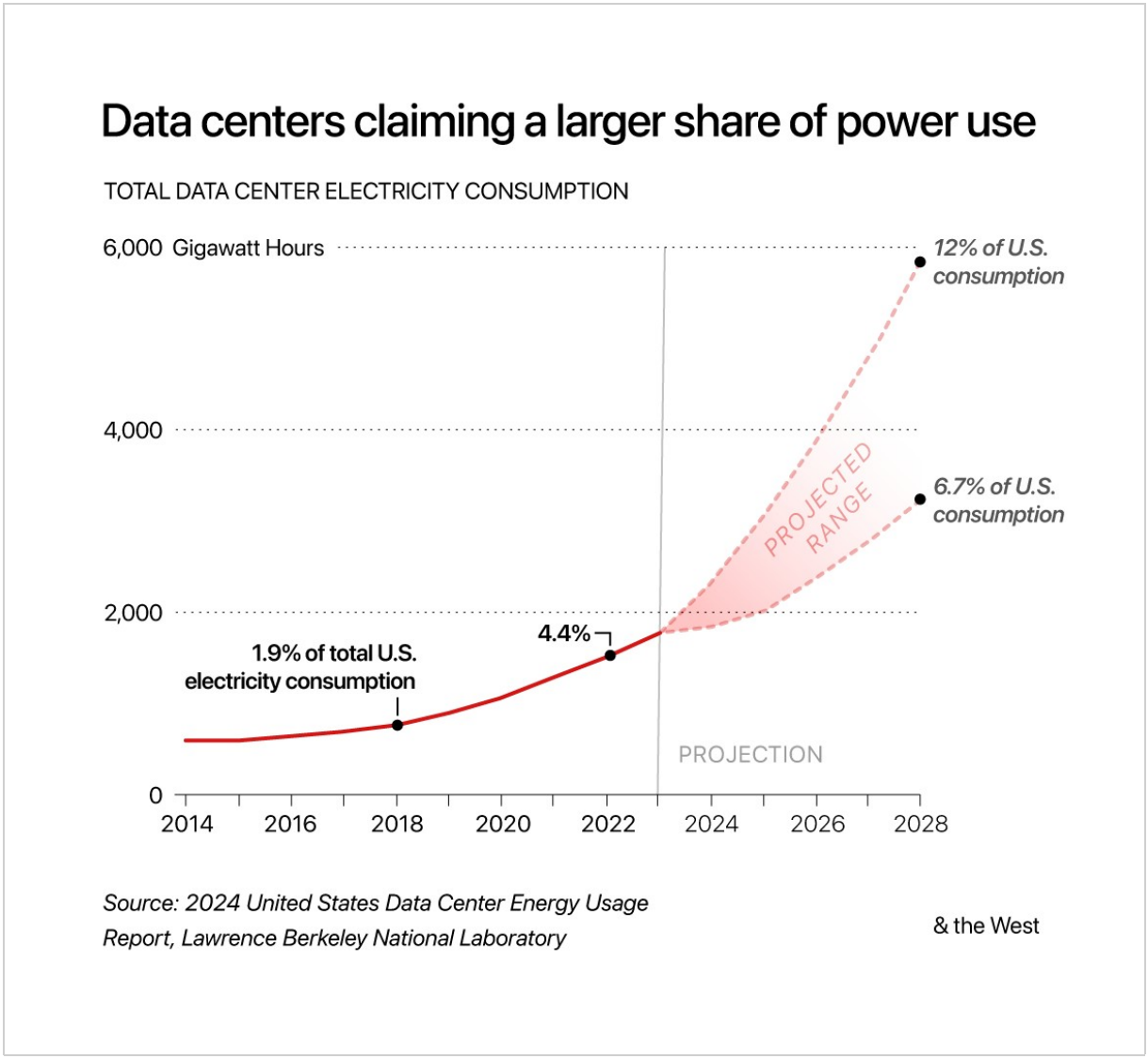
Data centers can be an awkward fit. In an interview, Alan Stephenson, Phoenix's deputy city manager, noted, "a data center takes a lot of land but they don't provide enough jobs for the infrastructure investment..." The planners' goal "is balancing public infrastructure investment with jobs that are part of quality of life.... That just has challenges when things like this come in that are such a big entity that they change the dynamic of what you're trying to create."

Case in point: in the village of Laveen < https://www.google.com/maps/place/Laveen+Village,+Phoenix,+AZ,+USA/@33.3582961,-112.1758926,10459m/data=!3m2!1e3!4b1!4m6!3m5!1s0x872b10dc74a5df79:0x35df73622f807954!8m2!3d33.3825705!4d-112.1236468!16s%2Fg%2F1tdfnqvq?entry=ttu&g_ep=EgoyMDI1MDMyNS4xIKXMDSoASAFQAw%3D%3D> , on the once-rural edge of the city, now houses 62,421 people, up from about 9,000 25 years ago. Planners spent years studying the transportation < https://www.phoenix.gov/pddsite/Documents/PZ/pdd_pz_pdf_00109.pdf> infrastructure needed to support the fast-growing local housing developments. Late last year, Amazon paid $277 million to buy < https://www.datacenterdynamics.com/en/news/amazon-acquires-220-acres-in-phoenix-arizona-for-possible-data-center/> 220 open acres in Laveen for a new data center. < https://www.datacentermap.com/usa/arizona/phoenix/amazon-aws-loop-202-dobbins-road/>

'Hyperscale' centers drive increasing power consumption

The new hyperscale data centers are a different animal from their predecessors. A decade ago, an average data center used 20 megawatt hours of power a month, about 20 times the use of

an average home < https://www.eia.gov/tools/faqs/faq.php?id=97&t=3> . The newest generation of large-scale data center campuses that support artificial intelligence with its large energy-consuming chips can use 100 or more megawatt hours a month, or five times their predecessors < https://www.statista.com/topics/13055/data-center-power/> . The Lawrence Berkeley National Laboratory's recent report said that data centers used 4.4 percent of the nation's electricity in 2022 < https://www.statista.com/topics/13055/data-center-power/> and would use 6.7 to 12 percent by 2028.



The Salt River Project, the nonprofit utility which provides electricity to 1.1 million customers in Phoenix, last year requested proposals < https://www.utilitydive.com/news/srp-seeks-1200-MW-peak-capacity-phoenix-growth/707633/> from energy companies that would let it increase its peak-load capacity by 1.2 gigawatts.

Karla Moran, an economic development manager at SRP, told Business Insider < https://www.businessinsider.com/data-centers-energy-demand-utilities-green-renewable-2023-10> 18 months ago, "we have 7,000 megawatts of data-center requests in

our pipeline." She **told Reuters < https://www.reuters.com/business/energy/rush-data-centers-creates-us-solar-hotspots-2024-02-22/>** that afternoon power consumption can go up five percent to accommodate both heating and cooling needs; an increase that doubles in the summer. Last August 4, **the second day in a row < https://media.srpnet.com/srp-sets-new-record-for-amount-of-energy-delivered-to-customers/>** of 116-degree heat in Phoenix, SRP delivered 8,219 megawatts of power, the most ever.

A former Arizona utility executive, speaking on background, added that the utility has 42-data-center projects on deck totaling more than 10,255 MW. While not all will be built, "you get a sense of the magnitude" of new demand. SRP "would have to more than double its current generation resource mix to serve that load."

Arizona's utilities have formed special teams to work with data centers and other customers requiring large loads. Data-center companies want to create new campuses rapidly and their timelines are out of sync with the time utilities need to assemble the needed generation, transmission and distribution infrastructure.

"The teams are trying to get an understanding of what [is needed] and by when," the former executive said. To ensure a fair allocation of costs, utilities require data centers and other big customers to pay upfront for the needed infrastructure. "They are trying to ensure that everybody pays their fair share," the executive concluded.

Oregon legislators are considering **a bill < https://www.opb.org/article/2025/03/06/power-act-data-centers-electricity-use-oregon-lawmakers-energy-google-utilities/>** that would ensure data centers pay their share of electricity bills. Pam Marsh, who sponsored the legislation, said in an interview, "To be clear on this data center thing – we are all also consumers of technology. There's no question that the data centers are producing a product and consumers are consuming it. But in the ratepayer regulatory world, it's important that we are separate in what ratepayers pay."

> "There's no question that the data centers are producing a product and consumers are consuming it. But in the ratepayer regulatory world, it's important that we are separate in what ratepayers pay."
>
> – Rep. Pam Marsh, Oregon state legislature

Two Harvard professors recently published a paper **"Extracting Profits from the Public: How Utility Ratepayers Are Paying for Big Tech's Power" < https://eelp.law.harvard.edu/wp-content/uploads/2025/03/Harvard-ELI-Extracting-Profits-from-the-Public.pdf>** questioning whether other rate-payers subsidize the centers. David Roberts, hosting the authors recently **on his podcast "Volts," < https://www.volts.wtf/p/who-is-paying-for-all-that-data-center?utm_source=podcast-**

email%2Csubstack&publication_id=193024&post_id=158496528&utm_campaign=email-play-on-substack&utm_medium=email&utm_content=play_card_play_button&r=zlfg&triedRedirect=true> said that setting rates among different types of users has always been problematic. "These are inherently subjective decisions," he said.

Ari Peskoe, one of the paper's authors, told Roberts, "there's certainly right now a nationwide competition to attract these data centers because building the infrastructure can be a sort of lucrative business for the utilities." Some utilities have created special contracts for data-center companies. This can lead to a realignment of all customers' rates. Because of their special deals, data-center companies may get a break, but it is difficult to tease out if they do.

In an article < https://www.tdworld.com/smart-utility/article/55138699/a-different-approach-to-data-center-demand> in T&D World last fall, Clift Pompée, a vice-president at Compass Datacenters in North Carolina agreed that, "shareholders and other ratepayers shouldn't be left holding the bag for million- or billion-dollar generation and transmission investments…." Centers should "chart out a reasonable multi-year load ramp to enable our utility partners to plan."

More companies seek to co-locate generating capacity near a data center. SRP is working with < https://www.datacenterdynamics.com/en/news/orsted-commissions-hybrid-solar-plant-in-arizona-tied-to-meta-data-center/> Meta and the Danish energy firm Ørsted to build a combination of a 300-MW solar field and a comparable battery array near Meta's new Mesa data-center campus, just east of Phoenix.

Growing demand strains transmission capacity

Data-center companies and utilities have to worry not just about how much power they need but how many new transmission lines will carry it. The Salt River Project announced last fall < https://media.srpnet.com/arizona-power-plant-and-transmission-line-siting-committee-hearings-for-srp-south-mountain-transmission-project-application-to-begin-nov-12/> that it has sent the Arizona Corporation Commission its recommendation for a new substation and two new 230 kV transmission lines – the South Mountain Transmission Project.

The company's 2023 Integrated Systems Plan < https://www.srpnet.com/assets/srpnet/pdf/grid-water-management/grid-management/isp/SRP-2023-Integrated-System-Plan-Report.pdf> said that "SRP will likely need to double or triple resource capacity at an unprecedented pace in the next decade" and that "hundreds of miles of new or upgraded transmission lines and nearly double the number of … transformers could be needed…"

In Santa Clara, the city council in November approved < https://www.svvoice.com/new-transmission-line-will-fuel-svp-growth/> a 2.24-mile new overhead transmission line.

Earlier in the fall, its utility, Silicon Valley Power issued **$143 million in revenue bonds < https://www.kuam.com/story/31135232/2016/02/Wednesday/new-bonds-anticipated-to-support-utility-upgrades>** to pay for upgrading the grid infrastructure. And it recently announced a five percent rate increase for all customers. To ensure that homeowners aren't footing the bill for data centers' power use, a California legislator has proposed a measure **requiring utilities to have a separate rate structure for data centers < https://www.publicpower.org/periodical/article/state-lawmakers-address-costs-and-other-issues-tied-data-centers>** .

In rural Oregon, the Umatilla Electric Company, which serves Morrow, Umatilla, and Union Counties, has 11,000 customers and 2,300 miles of transmission lines in its 1,900 square miles of service territory. Two local towns, Boardman and Hermiston, are nodes of data-center growth. The company is building a new **5.6 mile overhead 230-kilovolt power line. < https://www.umatillaelectric.com/mcnary-to-wanapa-transmission-line/>**

Some data center companies are exploring locating centers next to power sources, particularly small modular nuclear reactors. Oregon law makes it almost impossible to install such facilities, but **a bill in the legislature < https://olis.oregonlegislature.gov/liz/2025R1/Downloads/MeasureDocument/SB0996/Introduced>** this year could ease those restrictions.



A 2007 aerial view of the since-demolished Boardman Coal Plant in Hermiston County, owned by Portland General Electric. In the distance is the Columbia River. *Sam Beebe via* ***Flickr < https://flickr.com/photos/sbeebe/3718329744/in/photolist-6EzpTw-S7CF8c-21p3ZRF-RGzVHk-5ydUEg-sJAfMP-2gSP5Sd-5xpaBy-5xNHeM-5xitqF-GSrTR-2gSP5DY-***

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While less water intensive than agriculture, cooling demands considerable quantities

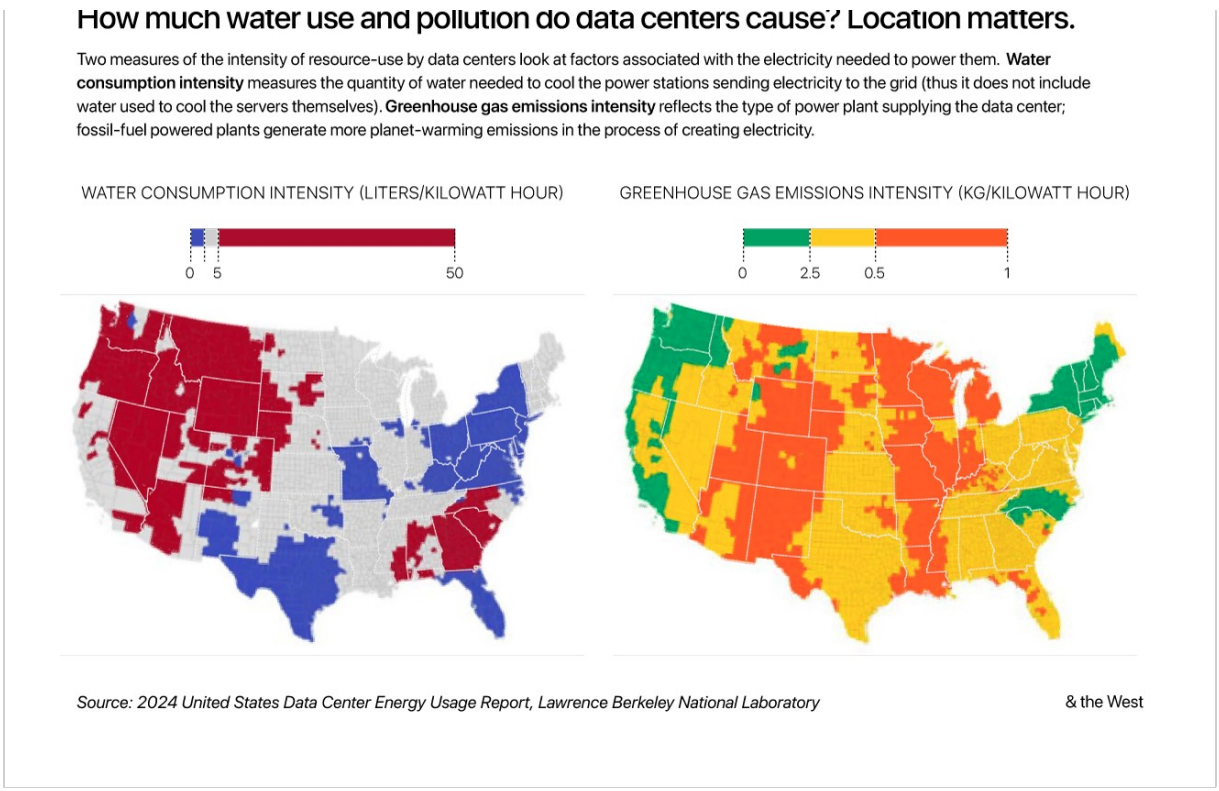
Two years ago **a report < https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/investing-in-the-rising-data-center-economy>** from McKinsey and Company, the management consulting firm, noted that "Data center equipment, often consisting of thousands of servers, must be cooled to work efficiently. Indeed, the capacity of a data center is dictated by how well it cools the servers—the more closely they can be stacked, the more productive the square footage."

> Water consumption by data centers is a serious concern, particularly in the desert Southwest.

Water consumption by data centers is a serious concern, particularly in the desert Southwest. Arizona has **limited home construction < https://www.nytimes.com/2023/06/01/climate/arizona-phoenix-permits-housing-water.html>** in the Phoenix area to preserve groundwater. Data centers' water consumption is significant. APM Research Lab recently **reported < https://www.apmresearchlab.org/10x/data-centers-resource#:~:text=An%20analysis%20Microsoft%20shared%20with,1.9%25%20of%20the%20town's%20population>** Meta's calculation that its center in Goodyear, west of Phoenix, used "around 56 million gallons of potable water annually, equivalent to 670 Goodyear households" or about two percent of the town's residences.

Of the three different cooling mechanisms often used to keep ambient temperatures from 68 to 82 degrees, two rely on large amounts of water, particularly evaporative cooling, which involves pushing hot air through pads soaked in water, which then evaporates and cools the air.

Masheika Allgood, an AI ethicist who consults with municipal governments, believes that so much water is evaporated for cooling that recycling can't replace what was lost. "When you are cooling a data center – a significant amount of the water is evaporated during the cooling process... Even if you don't evaporate all the water, you have to treat it. You can't recycle that ...." Water cooling methods, however, vary dramatically in how much water is consumed and some cooling methods use no water at all.



Water use can be a sensitive subject. Google, which operates Oregon data centers at The Dalles, a city of 16,000 people not far from Morrow County, resisted disclosing its water use, **paying $100,000 for the city's lawsuit < https://www.datacenterdynamics.com/en/news/we-now-know-how-much-water-googles-oregon-data-centers-use-after-city-drops-lawsuit-against-journalists/>** against The Oregonian, a Portland newspaper that had filed a public records request for the data. When the suit was dropped, Google's water-use totals were public: 355.1 million gallons, a quarter of the city's annual water use in 2021.

In Hermiston, as in Phoenix, agriculture's water use dwarfs that of data centers. In February, Hermiston and Amazon reached a 25-year agreement that would expand summertime water capacity by 100 million gallons annually for all users, thanks to a new way of storing excess winter flows. As the **Hermiston Herald reported < https://hermistonherald.com/2025/02/26/hermiston-makes-water-service-deal-with-amazon/>**, a pumping system would take some "400 million gallons of water from the Columbia River between October and March" when fish and farmers need less, and [store] it approximately 1,500 feet underground," making it available for all summertime users without new withdrawals.





Google's data center in The Dalles, Oregon, about four miles downstream from the Columbia River dam that provides hydropower to the region. *Tony Webster via Wikimedia Commons*

In an **interview with Oregon Public Broadcasting < https://www.opb.org/article/2023/05/19/amazon-data-center-oregon-morrow-county/>** two years ago, Morrow County's David Sykes said that during negotiations with Amazon, "We did talk water use ...because, especially here in Eastern Oregon, water is very important. There's industries that use a lot more water than they will. They use it for cooling in the summer. A lot of it's being recycled through. So, they're pretty good and judicious with their water use, we feel."

In a 2023 **paper < https://arxiv.org/pdf/2304.03271>** in the research-sharing platform ArXiv, four authors – three from the University of California-Riverside and one from the University of Texas at Arlington – wrote that worldwide, "Google's data center water consumption increased by [about] 20 percent from 2021 to 2022 and by [about]17 percent from 2022 to 2023 [8], and Microsoft saw [about] 34 percent and [about] 22 [percent] increases over the same periods..." They noted Energy Department estimates that by 2028, data centers' total annual on-site water consumption "could double or even quadruple the 2023 level."

An **Energy Department report < https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf>** noted that in 2014, U.S. data centers used 21.2 billion liters [5.6 billion gallons] of water. By 2023, 84 percent of the water was used to run the larger "hyperscale" centers, and the total water use had tripled to 66 billion liters [17.4 billion gallons].

A few months ago, Meta, in a ceremony as it began construction of its new data center in Mesa, Arizona, pledged that, **by 2030, it would restore more water to the region than it used < https://www.azcentral.com/story/money/business/tech/2025/01/31/meta-says-1-billion-data-center-in-mesa-is-sustainable/78031156007/>** , becoming "water-positive." Amazon Web Services made a similar **pledge < https://www.aboutamazon.com/news/aws/aws-water-positive-by-2030>** in 2022 for its network of data centers around the country.

Amazon, Meta and Apple are also **using reclaimed wastewater < https://datacentremagazine.com/articles/reclaimed-wastewater-to-be-used-at-20-aws-locations>** , not potable water, at many of their sites.

California's legislature is considering a measure that would require data-center companies applying for permits **to provide an estimate of expected water use < https://calmatters.digitaldemocracy.org/bills/ca_202520260ab93>** . Owners of existing centers would be required to report annual water use.

Carbon emissions are also a concern. In a recent **paper < https://arxiv.org/**

> The energy and environmental demands of the centers have increasingly been the focus of news coverage. Their role in enabling the internet-driven world people live in is often taken for granted.

html/2411.09786v1#:~:text=Our%20findings%20reveal%20that%20data,of%20US%20emissions%20in%202023).> , researchers from Harvard and the University of Pisa reported that "U.S. data centers produced 105 million tons CO2 equivalent gasses in the past year with a carbon intensity 48 percent higher than the national average." Data center emissions were about 2.18 percent of national carbon emissions.

Data centers' environmental impacts concern local officials. Santa Clara now requires new data centers to use carbon-free renewable electricity. Water-intensive once-through water cooling is also prohibited.

The energy and environmental demands of the centers have increasingly been the focus of news coverage. Their role in enabling the internet-driven world people live in is often taken for granted. But, as **Red Canary magazine reported < https://redcanarycollective.org/magazine/phoenix-data-center-water-usage/>** last month, "**thousands of small businesses wouldn't have survived the pandemic < https://www.axios.com/2021/06/04/pandemic-small-businesses-online>** if not for the e-commerce allowed by data centers. The research that both AI and data centers support through the storage and processing of data is also vital in tackling the climate crisis."

On a local level, enthusiasm is mixed.

In an interview, Santa Clara city councilman Sudhanshu "Suds" Jain said that he sees communities around the world racing with each other to attract data centers, and doesn't want his city to lose out. "They will go somewhere if we don't build them here....They are hugely beneficial financially to the city. They don't require security – they have their own. With few employees there is little demand on our resources. ... They don't impact the traffic."

But at the December council meeting where the latest center was approved, the planning commission chair, Salame, said, **according to the Silicon Valley Voice < https://www.svvoice.com/another-data-center-coming-to-santa-clara/>** , "An individual data center is not a problem, or two or ten, but we are at 60.... Do we want paved over data centers, edge to edge, or do we want to inspire other industries and residents and other aspects that can make the city more rich and more varied? Because, right now, I think we have reached a limit."

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Edited by Geoff McGhee.

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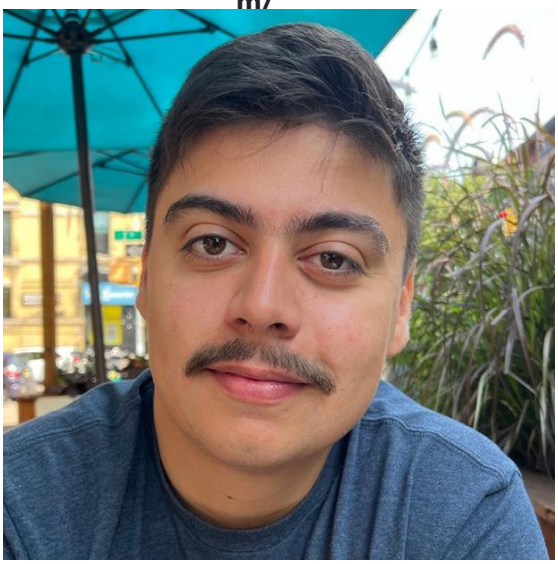
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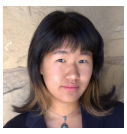
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